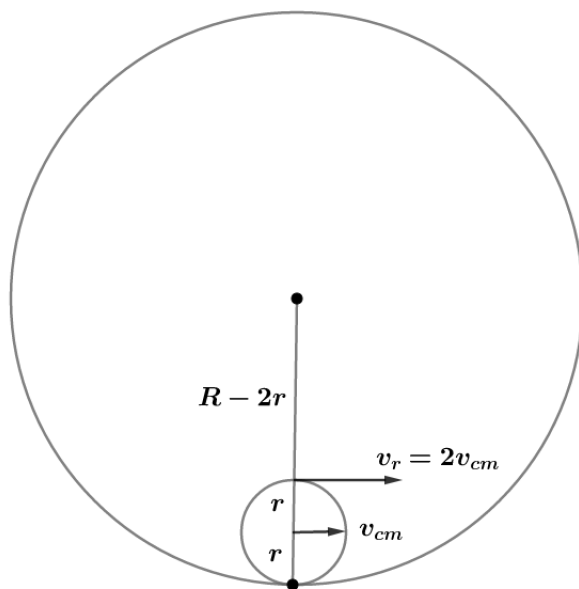


2018A F=ma Exam: Problem 8

Kevin S. Huang



Since the disk is rolling without slipping, we have

$$v_r = 2v_{cm}$$

The center of the disk travels distance $2\pi(R - r)$ in time T so

$$v_{cm} = \frac{2\pi(R - r)}{T}$$

Therefore,

$$v_r = \frac{4\pi(R - r)}{T}$$

so the answer is D.